NAME:
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First Solution:
Second Solution:

1. A stock price is currently $800. Over each of the next two 6-month periods it is expected to go up by 25% or down by 20%. The risk-free interest rate is 10% per annum with continuous compounding. Suppose a trader sells 2000 1-year ATM European put options (20 contracts, each contract is for 100 stocks). What position in the stock is necessary to hedge the trader's position at the time of the trade?

a) 707 (long) position in the underlying asset
b) 707 (short) position in the underlying asset
c) 1293 (long) position in the underlying asset
d) 1293 (short) position in the underlying asset

2. A stock price is currently $800. Over each of the next two 6-month periods it is expected to go up by 25% or down by 20%. The risk-free interest rate is 10% per annum with continuous compounding. Suppose a trader buys 2000 1-year ATM European put options (20 contracts, each contract is for 100 stocks). A trader hedge the portfolio delta to zero in every time period. What position in the stock is necessary to hold if the the underlying asset price is decreasing during the first 6-month?

a) 707 (long) position in the underlying asset
b) 1293 (short) position in the underlying asset
c) 0 position in the underlying asset
d) 2000 (short) position in the underlying asset